

GSERM - St. Gallen 2026

Analyzing Panel Data

June 22, 2026

Logistics:

- Instructor: Prof. Christopher Zorn
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 - Bluesky / Instagram / etc.: [@prisonrodeo](#)
- Class: June 22-26, 2026, 09:15-15:15 CET.
- The course outline / syllabus is [here](#).
- More important: The syllabus, slides, readings, code, data, etc. are all available on the course [github repo](#) (viewable at <https://github.com/PrisonRodeo/GSERM-Panel-2026>).


GSERM-Panel-2026
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 .gitattributes	Initial commit	2 weeks ago
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 GSERM-2026-Useful-R-Resources.pdf	Useful R Resources	2 weeks ago
 GSERM-Panel-WDI-Description-2026.pdf	WDI Description	2 weeks ago
 GSERM-St-Gallen-Analyzing-Panel-Data...	Syllabus	2 weeks ago
 README.md	READ ME	2 weeks ago

About

Github repository for Analyzing Panel Data, Summer 2026

 Readme

 Activity

 0 stars

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 0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Evaluation at GSERM isn't easy... the plan:

- One “homework exercise”
 - Practical exercise – “real” data analysis and discussion
 - Assigned Tuesday (June 23); due Friday (June 26)
 - Worth 300 possible points
- Final Examination
 - Multiple essay-style questions + “real” data analysis
 - Some choice of questions to answer
 - Assigned Friday (June 26) at noon
 - Due either Friday, June 26, 2026 at 6:00 p.m. (“in-class” alternative) or Friday, July 3, 2026 (“take-home” alternative)
 - Worth 700 possible points
- Total course = 1000 possible points
- Grades assessed on Swiss (1 - 6) scale

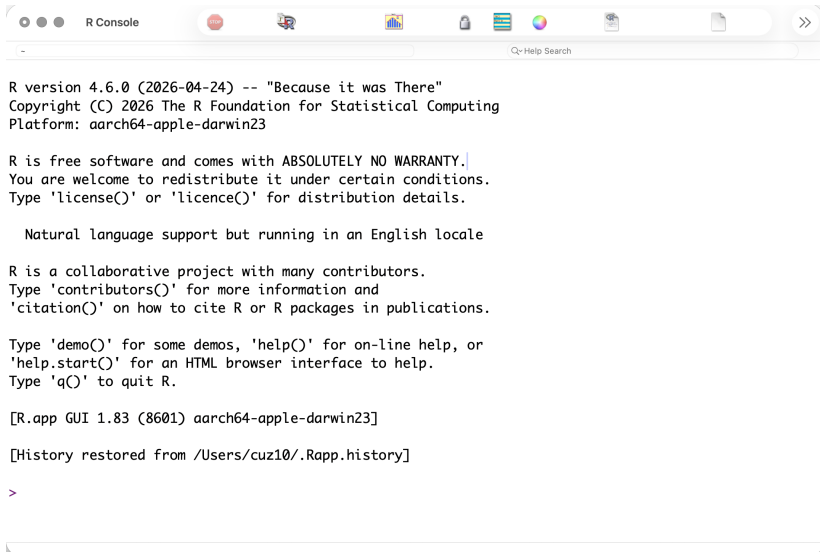
R

- All examples, plots, etc. are generated using R
- Current version (as of a week or so ago) is 4.6.0
- Desktop: Be sure to get the [RStudio / Posit](#) IDE...
- Alternatively: Can be run in a browser, using [Posit Cloud](#)
- The course Github repo contains a bit of [introductory code](#) for people who may never have used R, and a list of [R resources](#).
- A few of the primary packages we'll use include:
 - `plm`
 - `lme4`
 - `gee`

See the [econometrics](#) R [task view](#) for more.

Stata

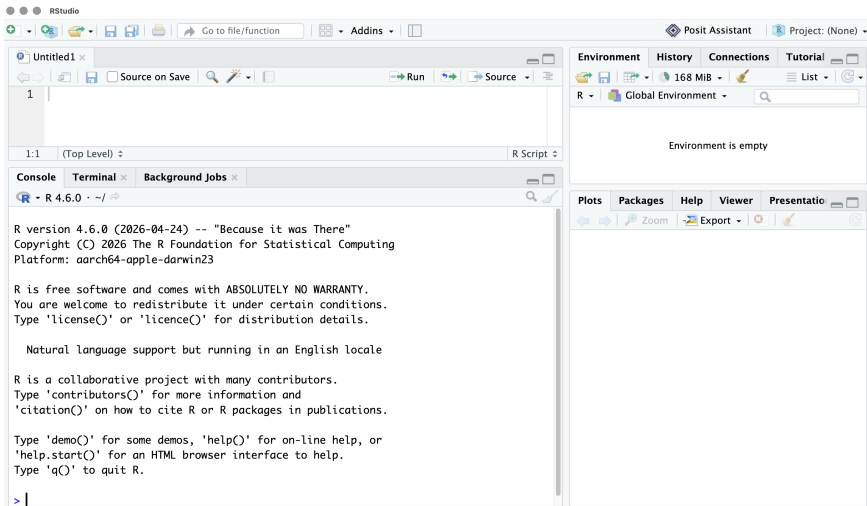
- Current version is v. 19
- Mostly use the `-xt-` series of commands (for “cross-sectional time series”)



The screenshot shows the R Console window with a title bar that says "R Console". The window contains the following text:

```
R version 4.6.0 (2026-04-24) -- "Because it was There"  
Copyright (C) 2026 The R Foundation for Statistical Computing  
Platform: aarch64-apple-darwin23  
  
R is free software and comes with ABSOLUTELY NO WARRANTY.  
You are welcome to redistribute it under certain conditions.  
Type 'license()' or 'licence()' for distribution details.  
  
Natural language support but running in an English locale  
  
R is a collaborative project with many contributors.  
Type 'contributors()' for more information and  
'citation()' on how to cite R or R packages in publications.  
  
Type 'demo()' for some demos, 'help()' for on-line help, or  
'help.start()' for an HTML browser interface to help.  
Type 'q()' to quit R.  
  
[R.app GUI 1.83 (8601) aarch64-apple-darwin23]  
  
[History restored from /Users/cuz10/.Rapp.history]  
  
>
```

RStudio / Posit



RStudio (annotated)

Source window: This is the "Source" window. It's the place where you'll type the code that will then be sent to R. It's basically a text editor. You can open text files of any kind here if you want. Files that appear here end in (and should be saved with) the extension ".R" (as in "MyCode.R"). You'll spend most of your time working here.

Environment window: This is the "Environment" window. It is where you can find all the various "objects" that you create, grouped by object type (data frames, lists, graphs, etc.). Environment is empty. There's also a "History" tab above; switching to that will show what has transpired in the Console window recently.

Console window: This is the "working directory." Anything you save will be saved here, unless you tell the program to save it somewhere else. Type 'demo()' for some demos, 'help()' for on-line help, or 'help.start()' for an HTML browser interface to help. Type 'q()' to quit R. This is the "Console." When you run the code in the Source window, the results that aren't graphics appear here.

Annotations:

- Click here to save your source code. Save often! (points to the Save icon in the Source window toolbar)
- Highlight text in the Source window, then click this button to "run" the code. (points to the Run button in the Source window toolbar)

Starting Points

- Panel data: Data comprising repeated observations over time on a set of cross-sectional units.
- Terminology:
 - “Unit” / “Units” / “Units of observation” / “Panels” = Things we observe repeatedly
 - “Observations” = Each (one) measurement of a unit
 - “Time points” = When each observation on a unit is made
 - $i \in \{1 \dots N\}$ indexes units
 - $t \in \{1 \dots T\}$ or $\{1 \dots T_i\}$ indexes observations / time points
 - If $T_i = T \forall i$ then we have “**balanced**” panels / units
 - Balanced panels also imply $N_t = N \forall t$
 - NT = Total number of observations (if balanced)
- “Panel” $\neq$ “Time Series”
- “Panel” $\neq$ “Multilevel” / “Hierarchical” / etc.

$N \gg T \rightarrow$ “panel” data...

- (American) National Election Study panel studies ($N \approx 2000$, $T = 3$)
- Swiss Household Panel (FORS) ($N = \text{large}$, $T = 26$)
- Often:
 - Cross-sectional units are a sample from a population
 - T is (relatively) fixed

$T \gg N$ or $T \approx N \rightarrow$ “time-series cross-sectional” (“TSCS”) data

- National OECD data ($N = 20$ original members, $T \approx 60$)
- Often:
 - N is an entire population, and is (relatively) fixed
 - Asymptotics are in T

$N = 1 \rightarrow$ “time series” data

$T = 1 \rightarrow$ “cross-sectional” data

Panel Data Structure + Organization

Typical: “long”:

id	t	Y	X	...
1	1	250	3.4	...
1	2	290	3.3	...
⋮	⋮	⋮	⋮	...
2	1	160	4.7	...
2	2	150	4.9	...
⋮	⋮	⋮	⋮	...

Sometimes: “wide”:

id	$Y1$	$Y2$	...	$X1$	$X2$	...
1	250	290	...	3.4	3.3	...
2	160	1250	...	4.7	4.9	...
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Introduction to Panel Variation: A Tiny (Fake) Example

Firm	Year	Sector	Inflation	FemaleCEO	NPM
1	2021	IT	0.6	0	21.6
1	2022	IT	2.8	0	19.9
1	2023	IT	2.1	0	23.1
1	2024	IT	1.1	0	22.0
2	2021	Energy	0.6	0	13.7
2	2022	Energy	2.8	0	12.1
2	2023	Energy	2.1	0	12.3
2	2024	Energy	1.1	0	13.0
3	2021	Retail	0.6	1	5.1
3	2022	Retail	2.8	1	4.9
3	2023	Retail	2.1	1	4.8
3	2024	Retail	1.1	0	5.0

Aggregation (means)

Cross-Sectional:

Firm	Year	Sector	Inflation	FemaleCEO	NPM
1	2022	IT	1.65	0.00	21.65
2	2022	Energy	1.65	0.00	12.78
3	2022	Retail	1.65	0.75	4.95

Temporal:

Year	Firm	Inflation	FemaleCEO	NPM
2021	2	0.6	0.333	13.5
2022	2	2.8	0.333	12.3
2023	2	2.1	0.333	13.4
2024	2	1.1	0.000	13.3

Aggregation:

- *Always* loses information
- Occasionally forces *arbitrary decisions* / transformations
- Sometimes *distorts* relationships

If you have variation in multiple dimensions, use it.

Two-Way Variation

Two “dimensions” of variation:

- Cross-sectional variation: how each unit is – on average – different from other units – a/k/a **between-unit** variation
- Temporal variation: how each measurement / time point is – on average – different from other time points on average – a/k/a/ **within-unit** variation

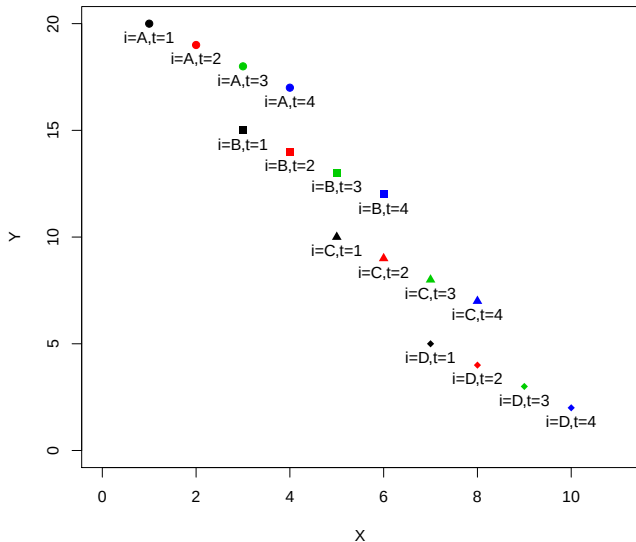
In panel data, a random variable may:

- Have only between-unit variation (i.e., lack *temporal variation*)
- Have only within-unit variation (i.e., lack *cross-sectional variation*)
- Have *both* within- and between-unit variation

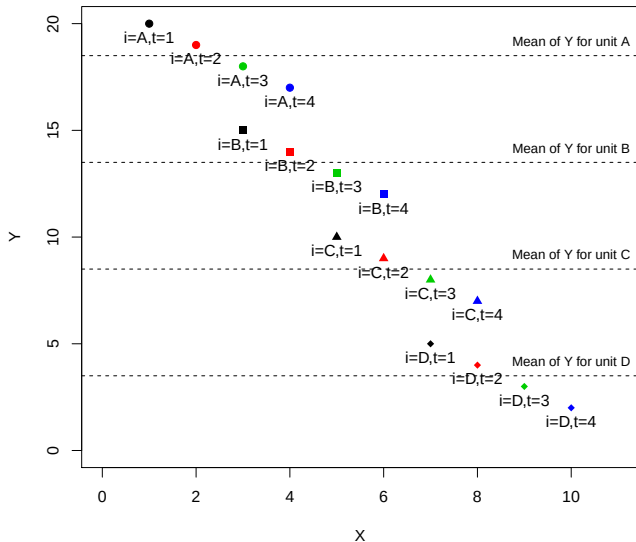
For Y_{it} , a variable that varies over both units and time:

- $\bar{Y}_i = \frac{1}{T_i} \sum_{t=1}^T Y_{it}$ is the over-time mean of Y for unit i ,
- $\bar{Y}_t = \frac{1}{N_t} \sum_{i=1}^N Y_{it}$ is the across-unit mean of Y at time t , and
- $\bar{Y} = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T Y_{it}$ is the grand mean of Y .

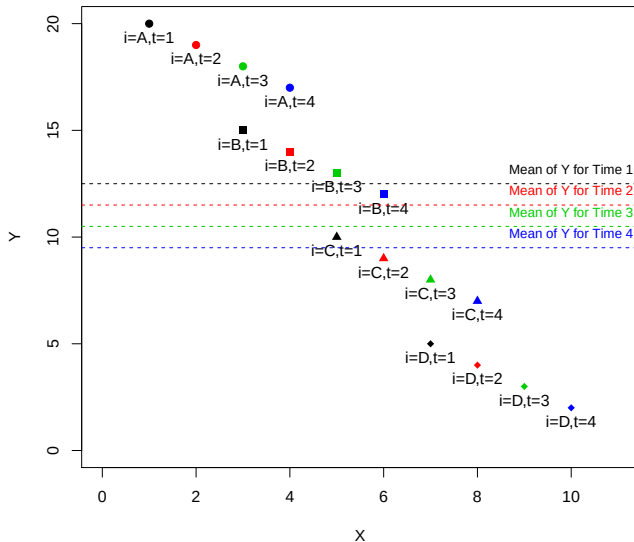
Dimensions of Variation



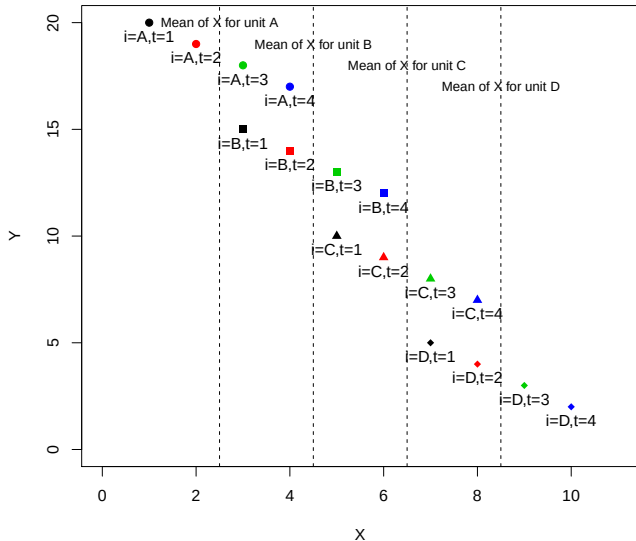
Dimensions of Variation: Y



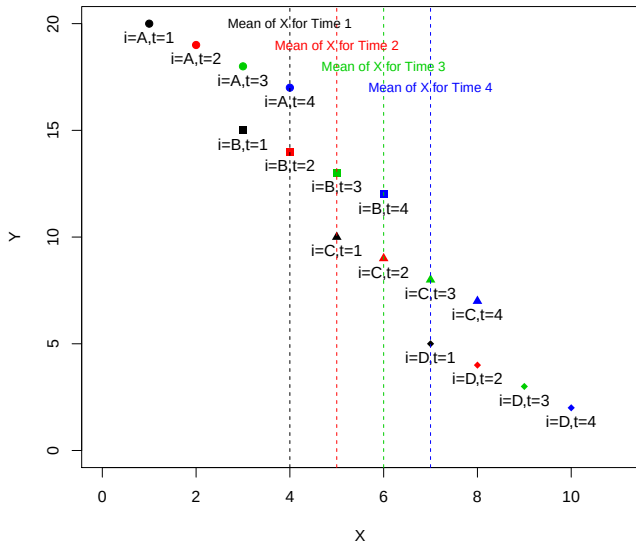
Dimensions of Variation: Y



Dimensions of Variation: X



Dimensions of Variation: X



Within- and Between-Unit Variation

The *within-unit* mean of Y is:

$$\bar{Y}_i = \frac{1}{T_i} \sum_{t=1}^{T_i} Y_{it}$$

That means that we can write:

$$Y_{it} = \bar{Y}_i + (Y_{it} - \bar{Y}_i).$$

That is, the *total* variation in Y_{it} can be decomposed into:

- The *between-unit* variation in the $\bar{Y}_i$ s, and
- The *within-unit* variation around $\bar{Y}_i$ (that is, $Y_{it} - \bar{Y}_i$).

Within- and Between-Time Point Variation

Note that (while unusual) one could do a similar decomposition vis-à-vis time:

$$Y_{it} = \bar{Y}_t + (Y_{it} - \bar{Y}_t).$$

That is, the *total* variation in Y_{it} can be decomposed into:

- The *temporal* variation in the $\bar{Y}_t$ s, and
- The *within-time-point* variation around $\bar{Y}_t$ (that is, $Y_{it} - \bar{Y}_t$).

In a similar fashion, we can also calculate the within- and between-unit variability (e.g., the standard deviations) of the constituent variables $\bar{Y}_i$ and $(Y_{it} - \bar{Y}_i)$...

Variation (“Toy” Data from Above, Y)

“Total” Variation:

```
> with(toy, describe(Y))
  vars  n mean  sd median trimmed mad min max range skew kurtosis  se
X1     1 16   11 5.9     11      11 7.4  2  20   18    0    -1.5 1.5
```

“Between” Variation:

```
> Ymeans <- ddply(toy, .(ID), summarise, Y=mean(Y))
> with(Ymeans, describe(Y)) # between-unit variation
  vars n mean  sd median trimmed mad min max range skew kurtosis  se
X1     1 4   11 6.5     11      11 7.4 3.5  18   15    0    -2.1 3.2
```

“Within” Variation:

```
> toy <- ddply(toy, .(ID), mutate, Ymean=mean(Y))
> toy$within <- with(toy, Y-Ymean)
> with(toy, describe(within)) # within-unit variation
  vars  n mean  sd median trimmed mad  min max range skew kurtosis  se
X1     1 16   0 1.1     0      0 1.5 -1.5 1.5    3    0    -1.6 0.29
```

Cross-sectional regression:

$$\underset{N \times 1}{\mathbf{Y}_i} = \underset{N \times K}{\mathbf{X}_i} \underset{K \times 1}{\boldsymbol{\beta}} + \underset{N \times 1}{\mathbf{u}_i}$$

...requires all the usual OLS assumptions:

- $E(\mathbf{u}) = 0$
- $E(\mathbf{u}\mathbf{u}') = \sigma^2 \mathbf{I}$
- $\text{Cov}(\mathbf{X}, \mathbf{u}) = 0$
- $\text{Rank}(\mathbf{X}) = K$

In addition, we usually assume:

- $\boldsymbol{\beta}_i = \boldsymbol{\beta} \forall i$

Regression with Panel Data

Key point: For the model

$$\mathbf{Y}_{it} = \mathbf{X}_{it}\boldsymbol{\beta} + \mathbf{u}_{it}$$

...the same is true.

Variable Intercepts

Unit-specific intercepts:

$$Y_{it} = \beta_{0i} + \beta_1 X_{it} + u_{it} \quad (1)$$

Time-point-specific intercepts:

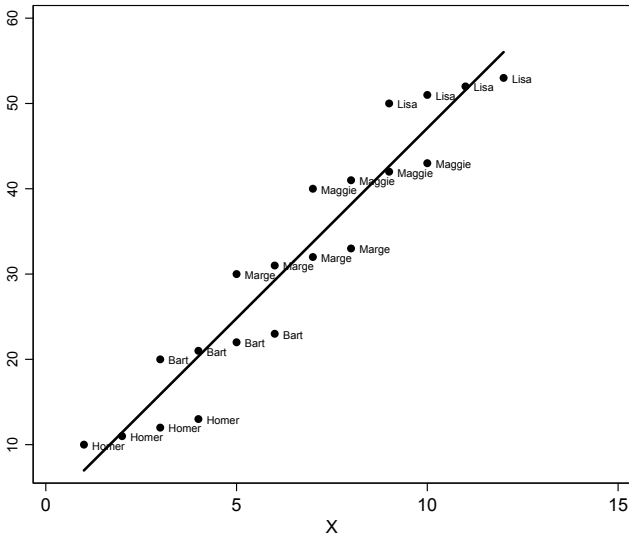
$$Y_{it} = \beta_{0t} + \beta_1 X_{it} + u_{it} \quad (2)$$

Both:

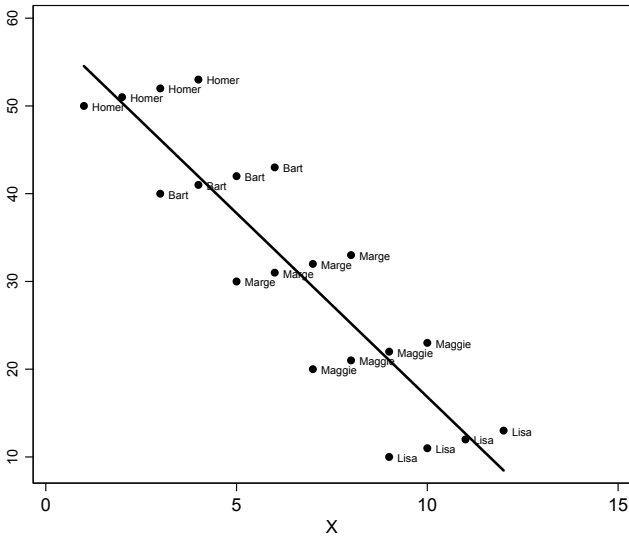
$$Y_{it} = \beta_{0it} + \beta_1 X_{it} + u_{it} \quad (3)$$

Note: Equation 3 is not identified (as written)!

Varying Intercepts



Varying Intercepts



Varying Slopes (+ Intercepts)

Unit-specific slopes:

$$Y_{it} = \beta_0 + \beta_{1i}X_{it} + u_{it} \quad (4)$$

(...one can also have time-point specific slopes, or both – again, the last of those is not identified as written.)

Unit-specific slopes + intercepts:

$$Y_{it} = \beta_{0i} + \beta_{1i}X_{it} + u_{it} \quad (5)$$

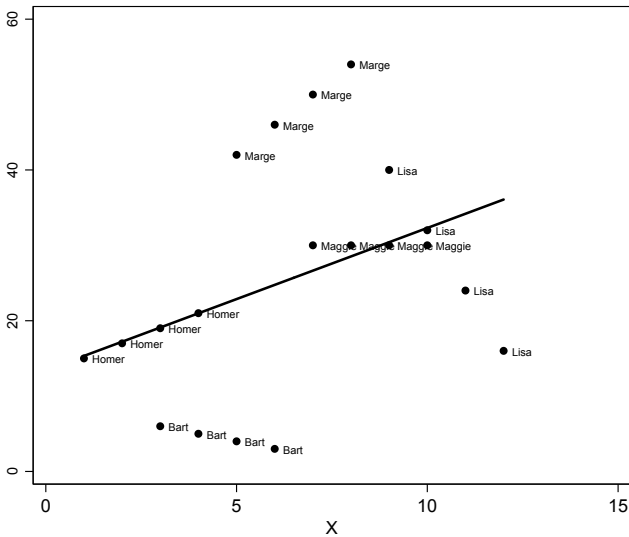
Time-point-specific slopes + intercepts:

$$Y_{it} = \beta_{0t} + \beta_{1t}X_{it} + u_{it} \quad (6)$$

Both...:

$$Y_{it} = \beta_{0it} + \beta_{1it}X_{it} + u_{it} \quad (7)$$

Varying Slopes + Intercepts



The Error Term...

Usual OLS assumption:

$$u_{it} \sim \text{i.i.d.} N(0, \sigma^2) \forall i, t$$

or, equivalently:

$$\mathbf{u}\mathbf{u}' \sim \sigma^2 \mathbf{I}$$

implies:

$$\text{Var}(u_{it}) = \text{Var}(u_{jt}) \forall i \neq j \text{ (i.e., no cross-unit heteroscedasticity)}$$

$$\text{Var}(u_{it}) = \text{Var}(u_{is}) \forall t \neq s \text{ (i.e., no temporal heteroscedasticity)}$$

$$\text{Cov}(u_{it}, u_{js}) = 0 \forall i \neq j, \forall t \neq s \text{ (i.e., no auto- or spatial correlation)}$$

Pooling = combining (repeated) observations on different units, and/or observations on different time points, into a single data frame.

Why should we pool data?:

- Adds data ($\rightarrow$ increases *precision*)
- Enhances *generalizability*

Every panel dataset requires that we make decisions about pooling.

Fitting (say) the model:

$$Y_{it} = \beta_0 + \beta_1 X_{it} + u_{it}$$

implies:

- that the process governing the relationship between X and Y is exactly the same for each i ,
- that the process governing the relationship between X and Y is the same for all t ,
- that the process governing the us is the same $\forall i$ and t as well.

Q: When can we “pool” data on different units?

“Partial” Pooling (Bartels 1996)

Two regimes:

$$Y_A = \beta'_A \mathbf{X}_A + u_A$$

$$Y_B = \beta'_B \mathbf{X}_B + u_B$$

with $\sigma_A^2 = \sigma_B^2$, and $\text{Cov}(u_A, u_B) = 0$.

Estimators:

$$\hat{\beta}_{A,B} = (\mathbf{X}'_{A,B} \mathbf{X}_{A,B})^{-1} \mathbf{X}'_{A,B} Y_{A,B}$$

and

$$\widehat{\text{Var}}(\beta_{A,B}) = \hat{\sigma}_{A,B}^2 (\mathbf{X}'_{A,B} \mathbf{X}_{A,B})^{-1},$$

A Pooled Estimator

Pooling A s and B s gives:

$$\begin{aligned}\hat{\beta}_P &= (\mathbf{X}'_A \mathbf{X}_A + \mathbf{X}'_B \mathbf{X}_B)^{-1} (\mathbf{X}'_A Y_A + \mathbf{X}'_B Y_B) \\ &= (\mathbf{X}'_A \mathbf{X}_A + \mathbf{X}'_B \mathbf{X}_B)^{-1} [\beta_A (\mathbf{X}'_A \mathbf{X}_A) + \beta_B (\mathbf{X}'_B \mathbf{X}_B)],\end{aligned}$$

What is the expectation?

$$\begin{aligned}E(\hat{\beta}_P) &= \beta_A + (\mathbf{X}'_A \mathbf{X}_A + \mathbf{X}'_B \mathbf{X}_B)^{-1} \mathbf{X}'_B \mathbf{X}_B (\beta_B - \beta_A) \\ &= \beta_B + (\mathbf{X}'_A \mathbf{X}_A + \mathbf{X}'_B \mathbf{X}_B)^{-1} \mathbf{X}'_A \mathbf{X}_A (\beta_A - \beta_B)\end{aligned}$$

We can assess whether $\hat{\beta}_A = \hat{\beta}_B$ via:

$$F = \frac{\frac{\hat{\mathbf{u}}_P' \hat{\mathbf{u}}_P - (\hat{\mathbf{u}}_A' \hat{\mathbf{u}}_A + \hat{\mathbf{u}}_B' \hat{\mathbf{u}}_B)}{K}}{\frac{(\hat{\mathbf{u}}_A' \hat{\mathbf{u}}_A + \hat{\mathbf{u}}_B' \hat{\mathbf{u}}_B)}{(N_A + N_B - 2K)}} \sim F_{[K, (N_A + N_B - 2K)]}$$

Bartels suggests:

$$\hat{\beta}_{\lambda} = (\lambda^2 \mathbf{X}'_A \mathbf{X}_A + \mathbf{X}'_B \mathbf{X}_B)^{-1} (\lambda^2 \mathbf{X}'_A Y_A + \mathbf{X}'_B Y_B)$$

with $\lambda \in [0, 1]$:

- $\lambda = 0 \rightarrow$ separate estimators for $\hat{\beta}_A$ and $\hat{\beta}_B$,
- $\lambda = 1 \rightarrow$ “fully pooled” estimator $\hat{\beta}_P$,
- $0 < \lambda < 1 \rightarrow$ a regression where data in regime A are given some “partial” weighting in their contribution towards an estimate of β .

“(R)oughly speaking, it makes sense to pool disparate observations if the underlying parameters governing those observations are sufficiently similar, but not otherwise.”

- Bartels (1996)

Our “Running Example” Data: WDI, 1960-2025

The World Development Indicators

- Cross-national country-level panel data
- $N = 215$ countries, $T = 66$ years (1960-2025) + missingness
- Variable types:
 - Geography: land area, arable land
 - Population indicators
 - Demographics: Birth rates, life expectancy, etc.
 - Economics: GDP, inflation, trade, FDI, etc.
 - Governments: expenditures, policies, etc.
- Full descriptions are listed in the Github repo [here](#).

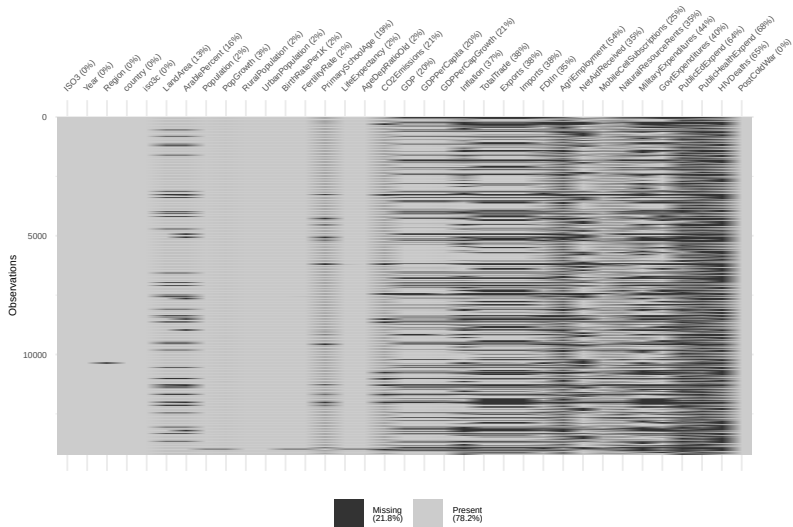
Summary Statistics: WDI Data

```
> describe(wdi,fast=TRUE,ranges=FALSE,check=TRUE)
```

	vars	n	mean	sd	skew	kurtosis	se
ISO3	1	14190	NaN	NA	NA	NA	NA
Year	2	14190	1992.50	19.05	0.00	-1.20	0.16
Region	3	14124	NaN	NA	NA	NA	NA
country	4	14190	NaN	NA	NA	NA	NA
iso3c	5	14190	NaN	NA	NA	NA	NA
LandArea	6	12288	596587.32	1645011.14	5.46	34.81	14839.81
ArablePercent	7	11871	13.40	13.68	1.47	1.96	0.13
Population	8	13945	25537942.47	106126340.56	9.72	105.61	898698.32
PopGrowth	9	13728	1.74	1.80	0.73	21.23	0.02
RuralPopulation	10	13975	47.79	25.80	-0.09	-0.99	0.22
UrbanPopulation	11	13975	52.21	25.80	0.09	-0.99	0.22
BirthRatePer1K	12	13945	27.51	13.19	0.25	-1.23	0.11
FertilityRate	13	13943	3.80	1.99	0.45	-1.17	0.02
PrimarySchoolAge	14	11539	6.13	0.61	-0.03	0.13	0.01
LifeExpectancy	15	13941	65.31	11.25	-0.77	0.19	0.10
AgeDepRatioOld	16	13945	10.84	7.27	1.80	5.05	0.06
CO2Emissions	17	11165	5.04	11.35	8.69	102.26	0.11
GDP	18	11307	243922334825.73	1153249695108.14	11.54	161.13	10845501840.29
GDPPerCapita	19	11307	12443.80	19449.79	3.17	15.21	182.91
GDPPerCapGrowth	20	11184	1.91	6.54	1.68	40.87	0.06
Inflation	21	8949	22.85	319.71	54.29	3652.65	3.38
TotalTrade	22	8802	78.15	53.67	3.01	17.96	0.57
Exports	23	8802	36.40	28.77	2.98	16.46	0.31
Imports	24	8811	41.76	27.49	2.52	13.58	0.29
FDIIn	25	9198	5.46	44.41	15.43	561.70	0.46
AgriEmployment	26	6496	27.25	23.03	0.68	-0.64	0.29
NetAidReceived	27	9218	528849286.43	1113709450.10	11.21	269.53	11599881.51
MobileCellSubscriptions	28	10587	39.15	53.62	1.26	1.61	0.52
NaturalResourceRents	29	9211	6.84	11.06	2.60	8.04	0.12
MilitaryExpenditures	30	7892	2.68	3.18	9.43	238.04	0.04
GovtExpenditures	31	8470	16.39	8.26	3.65	32.91	0.09
PublicEdExpend	32	5125	4.29	2.09	6.10	158.28	0.03
PublicHealthExpend	33	4547	3.31	2.40	1.37	3.43	0.04
HIVDeaths	34	5020	7002.28	20832.84	5.74	45.55	294.03
PostColdWar	35	14190	0.55	0.50	-0.18	-1.97	0.00

Missing Data

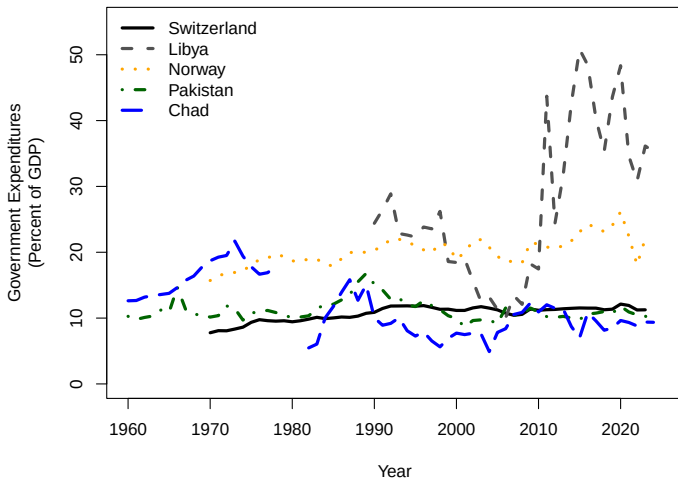
```
> library(naniar)
```



Exploring Variation: GovtExpenditures

General Government Final Consumption Expenditure (% of GDP)

"General government final consumption expenditure (formerly general government consumption) includes all government current expenditures for purchases of goods and services (including compensation of employees). It also includes most expenditures on national defense and security, but excludes government military expenditures that are part of government capital formation."



Panel Summary Statistics: GovtExpenditures

Summary: Government Expenditures

Variable	Dim	Mean	SD	Min	Max	Observations
GovtExpenditures	overall	16.391	8.255	1.788	147.735	N = 8470
	between		9.619	2.26	78.532	n = 185
	within		4.34	-33.947	85.593	T = 45.784

This means:

- The overall average of GovtExpenditures is 16.4
- The overall standard deviation of GovtExpenditures is 8.3
- The between-country standard deviation (standard deviation of the 185 country-level means) of GovtExpenditures is 9.6
- The within-country standard deviation of GovtExpenditures is 4.3
- GovtExpenditures is *cross-sectionally dominated*

Exploring Variation: Inflation

Inflation, Consumer Prices (Annual %)

"Inflation as measured by the consumer price index reflects the annual percentage change in the cost to the average consumer of acquiring a basket of goods and services that may be fixed or changed at specified intervals, such as yearly. The Laspeyres formula is generally used."

Summary: (logged) Inflation¹

Variable	Dim	Mean	SD	Min	Max	Observations
ln(Inflation)	overall	1.663	1.288	-6.909	10.076	N = 8406
	between		0.801	-1.2	4.243	n = 192
	within		1.092	-6.222	8.857	T = 43.781

The point: Inflation is *temporally dominated*.

¹Note: Logging inflation leads to missing values in the (relatively rare) instances where deflation occurs.

Exploring Variation: MidEast

Create a variable that = 1 if the country is in the Middle East, 0 otherwise:

```
> wdi$MidEast<-ifelse(wdi$Region=="Middle East & North Africa",1,0)
```

Summary: Mid-East

Variable	Dim	Mean	SD	Min	Max	Observations
MidEast	overall	0.098	0.298	0	1	N = 14124
	between		0.298	0	1	n = 214
	within		0	0.098	0.098	T = 66

Note:

- Approximately 9.8 percent of all the country-years in the data are for countries defined as being in the Middle East
- MidEast has *no* within-country variation, so...
- ...it has the same value for a particular country for every year in which that country is in the data.

Exploring Variation: PostColdWar

The PostColdWar variable is coded:

- 0 if that observation occurred ≤ 1989
- 1 if that observation occurred ≥ 1990

Summary: Post-Cold War

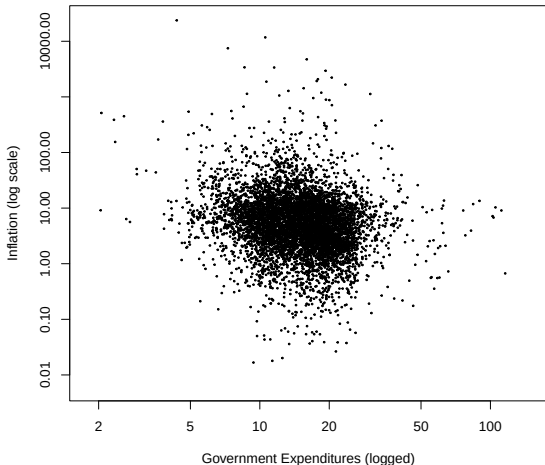
Variable	Dim	Mean	SD	Min	Max	Observations
PostColdWar	overall	0.545	0.498	0	1	N = 14190
	between		0	0.545	0.545	n = 215
	within		0.498	0	1	T = 66

Note:

- Approximately 54 percent of all the country-years in the data occur after the end of the Cold War
- PostColdWar has *no* between-country variation, so...
- ...in a given year, it has the same value for every country that is in the data that year.

Regression!

Q: What, if anything, is the (contemporaneous) relationship between (logged) inflation rates and (logged) government spending?



Inflation: Regression

Regression of lnInflation on (logged) GovtExpenditures: all /
pooled data:

```
> Inf.fit<-lm(lnInflation~lnGovtExp,data=wdi)
> summary(Inf.fit)
```

Call:

```
lm(formula = lnInflation ~ lnGovtExp, data = wdi)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.241	-0.689	-0.011	0.708	7.713

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.2526	0.1038	31.4	<2e-16 ***
lnGovtExp	-0.6038	0.0382	-15.8	<2e-16 ***

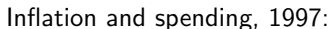
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.25 on 6623 degrees of freedom
(7565 observations deleted due to missingness)

Multiple R-squared: 0.0364, Adjusted R-squared: 0.0363

F-statistic: 250 on 1 and 6623 DF, p-value: <2e-16

Inflation and spending, 1997:



```
Call:
lm(formula = lnInflation ~ lnGovtExp, data = wdi97)
```

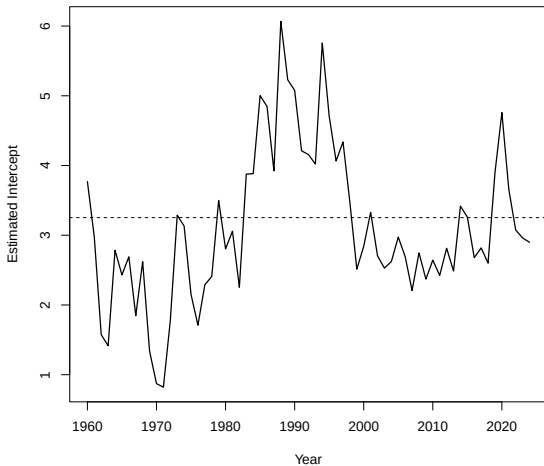
	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	4.338	0.732	5.93	0.000000026	***
lnGovtExp	-1.000	0.270	-3.70	0.00032	***

Residual standard error: 1.25 on 129 degrees of freedom
(84 observations deleted due to missingness)

F-statistic: 13.7 on 1 and 129 DF, p-value: 0.000316

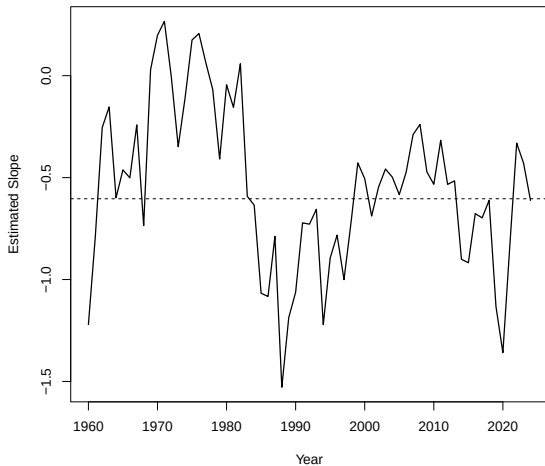
Intercepts For Every Year, 1960-2024

Estimated intercepts:



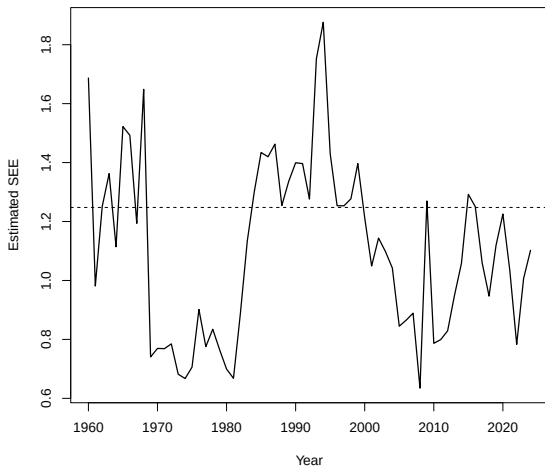
Slopes For Every Year, 1960-2024

Estimated slopes:

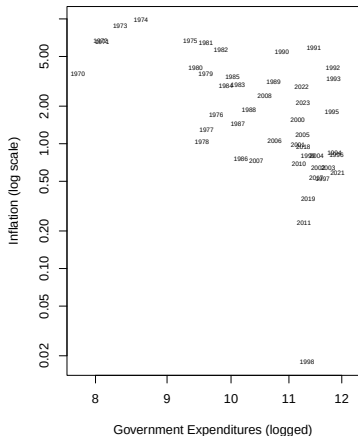


SEEs ($\hat{\sigma}$) For Every Year, 1960-2024

Estimated SEEs:



Time Series Regression: Switzerland



Inflation and spending in Switzerland:

```
> Inf.fitCHE<-lm(lnInflation~lnGovtExp,data=CHE)
> summary(Inf.fitCHE)
```

Call:

```
lm(formula = lnInflation ~ lnGovtExp, data = CHE)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.016	-0.373	0.017	0.579	1.781

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.84	2.90	4.76	0.000020 ***
lnGovtExp	-5.68	1.23	-4.60	0.000034 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

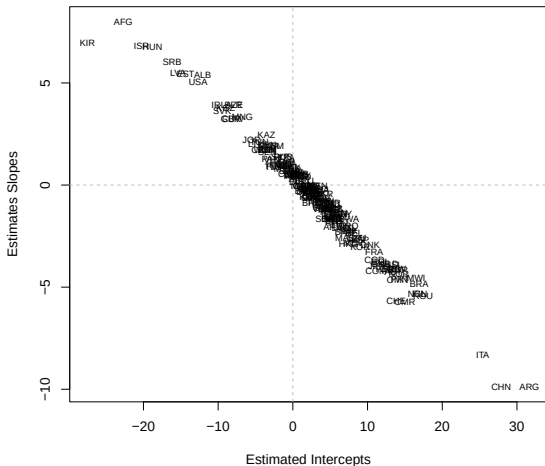
Residual standard error: 0.955 on 45 degrees of freedom
(19 observations deleted due to missingness)

Multiple R-squared: 0.32, Adjusted R-squared: 0.305

F-statistic: 21.2 on 1 and 45 DF, p-value: 0.0000343

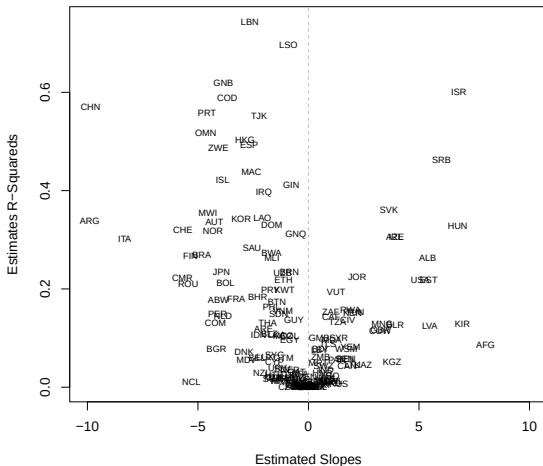
Country-Level Regressions: Intercepts & Slopes

Estimated intercepts and slopes:



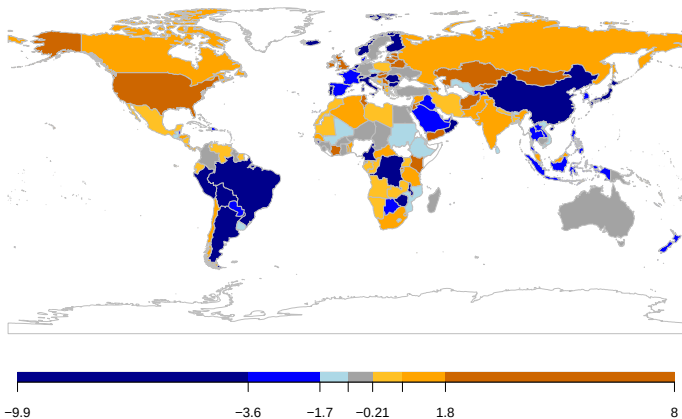
Country-Level Regressions: Slopes & R^2 s

Estimated slopes and R^2 s:



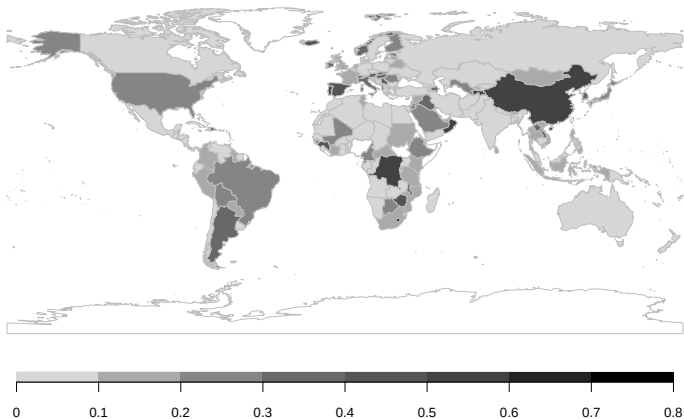
Maps! Country-Level Slopes ($\hat{\beta}_1$ s)

Estimated Slopes



Maps! Country-Level R^2 s

Estimated R-Squareds



An interactive tool for exploring panel data...

- Creator: Joachim Gassen (Department of Accounting, Humboldt-Universität zu Berlin)
- Built upon / consistent with `ggplot` / `tidyverse`
- Requires installing the ExPanDaR package
- Calling

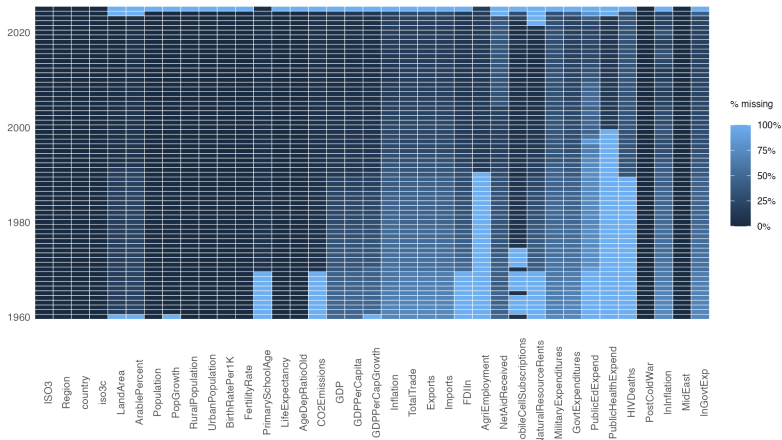
```
> ExPanD()
```


...opens the Shiny app, and asks for a (pre-formatted) data frame (typically in CSV format)
- More information is here: <https://github.com/trr266/ExPanDaR>

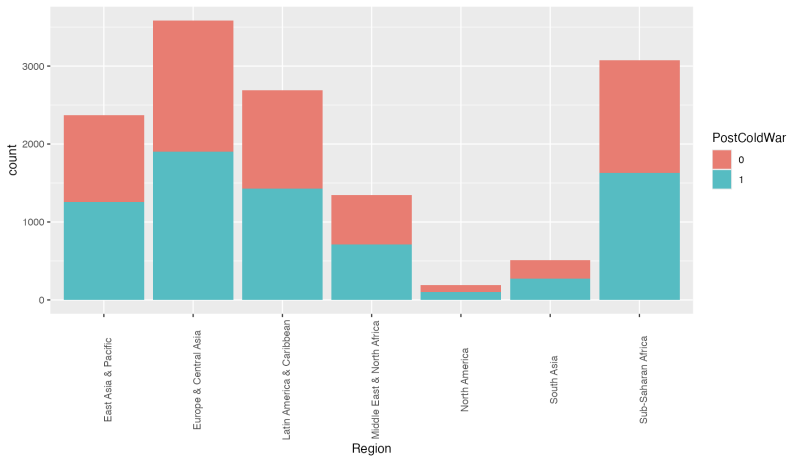
Some examples...

Missing Values

This graphs shows the ratio of missing values for all variable years.



Counts by factors:



ExPanDaR: More Summaries

Summary statistics:

Descriptive Statistics

Hover over variable names with mouse to see variable definitions.

Select Tab to choose the analysis set of variables or the base set of variables (to define new variables).

Click here to delete selected variables from the analysis sample.

Delete Variables

Click here to delete all user defined variables and to restore the original variable set of the analysis sample.

Restore Sample

Analysis Set		Base Set						
Variable	N	Mean	Std. dev.	Min.	25 %	Median	75 %	Max.
LandArea	12,288	596,587.319	1,645,011.136	2.027	10,120.000	100,540.000	469,930.000	16,389,950.000
ArablePercent	11,871	13.404	13.681	0.043	2.881	9.330	19.231	73.389
Population	13,945	25,537,942.467	106,126,340.562	2,715.000	512,554.000	4,330,364.000	14,215,611.000	1,450,935,791.000
PopGrowth	13,728	1.739	1.803	-27.471	0.657	1.671	2.660	21.700
RuralPopulation	13,975	47.791	25.796	0.000	27.515	47.878	68.794	98.161
UrbanPopulation	13,975	52.209	25.796	1.839	31.206	52.122	72.485	100.000
BirthRatePer1K	13,945	27.510	13.186	3.519	15.597	25.826	39.316	57.613
FertilityRate	13,943	3.798	1.990	0.582	2.010	3.221	5.663	8.864
PrimarySchoolAge	11,539	6.129	0.611	4.000	6.000	6.000	6.000	8.000
LifeExpectancy	13,941	65.308	11.247	10.989	58.561	67.945	73.443	86.497
AgeDepRatioOld	13,945	10.842	7.274	1.012	5.921	7.882	14.341	74.792
CO2Emissions	11,165	5.041	11.354	0.000	0.375	1.863	6.267	202.855
GDP	11,307	243,922,334,825.732	1,153,249,695,108.139	10,098,852.885	3,523,946,185.945	15,260,942,682.755	94,069,561,591.676	22,568,462,768,174.301
GDPPerCapita	11,307	12,443.796	19,449.795	122.679	1,353.546	3,997.788	16,081.178	247,170.220
GDPPerCapGrowth	11,184	1.908	6.541	-64.424	-0.498	2.101	4.514	140.491
Inflation	8,949	22.854	319.707	-17.640	2.135	4.822	10.070	23,773.132
TotalTrade	8,802	78.151	53.671	0.021	44.540	66.573	97.025	863.195
Exports	8,802	36.398	28.773	0.005	18.191	29.190	45.858	433.836
Imports	8,811	41.763	27.492	0.016	23.971	34.667	52.987	429.359
FDIn	9,198	5.457	44.406	-1,303.108	0.372	1.585	4.038	1,709.827
AgriEmployment	6,496	27.252	23.031	0.080	6.142	21.141	44.085	92.420

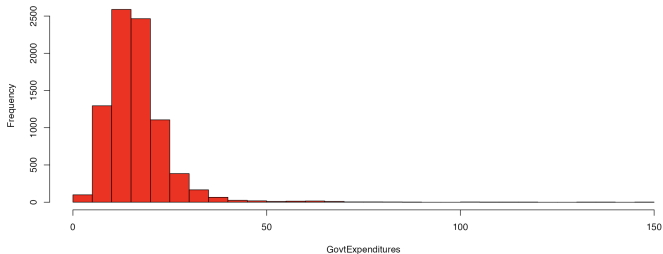
Histograms:

Histogram

Select variable to display

GovtExpenditures ▼

Suggested number of cells



Outlier Detection:

Extreme
ObservationsSelect variable to
sort data by

Inflation ▼

Select period to
subset to

All ▼

ISO3	Year	Inflation
COD	1994	23,773.1
BOL	1985	11,749.6
PER	1990	7,481.7
UKR	1993	4,734.9
AGO	1996	4,145.1
...	...	...
TCD	1986	-13.1
GNQ	1987	-13.2
IRQ	1996	-16.1
LSO	2009	-16.9
GNQ	1986	-17.6

Bar Charts of Means (by factors):

**By Group
Bar Chart**Select variable to
display

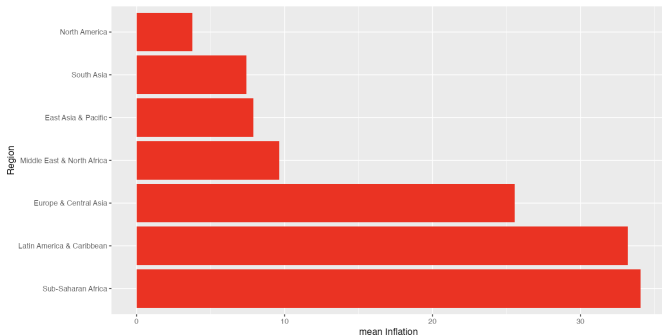
Inflation ▼

Select variable to
group by

Region ▼

Select statistic to
display

Mean ▼

☒ Sort by statistic

Violin Plots:

By Group Violin Chart

Select variable to
display

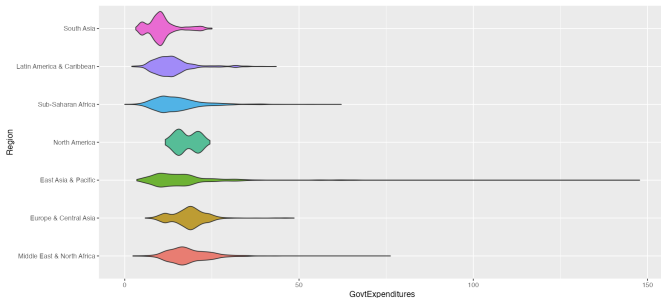
GovtExpenditures▼

Select variable to
group by

Region▼

☒ Sort by group means

(Note: Consider treating
your outliers if this
graph looks odd)



General Trends + Variation:

Time Trend Graph

Select first variable to display

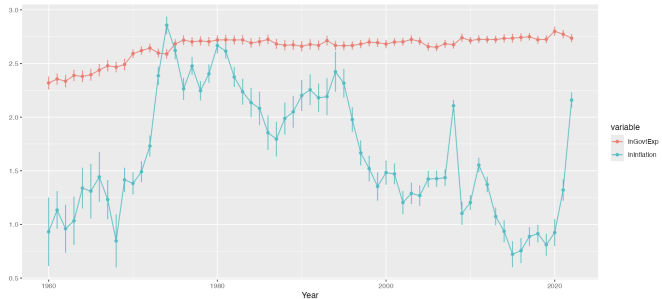
Inflation ▼

Select second variable to display

InGovtExp ▼

Select third variable to display

None ▼



ExPanDaR: More Trends

Trends in Quantiles:

Quantile Time Trend Graph

Select variable to display

Inflation

Quantiles to show:

- ☐ Min
- ☐ 1 %
- ☐ 5 %
- ☒ 10 %
- ☐ 25 %
- ☒ 50 %
- ☐ 75 %
- ☒ 90 %
- ☐ 95 %
- ☐ 99 %
- ☐ Max



ExPanDaR: Even More Trends

Trends By Group:

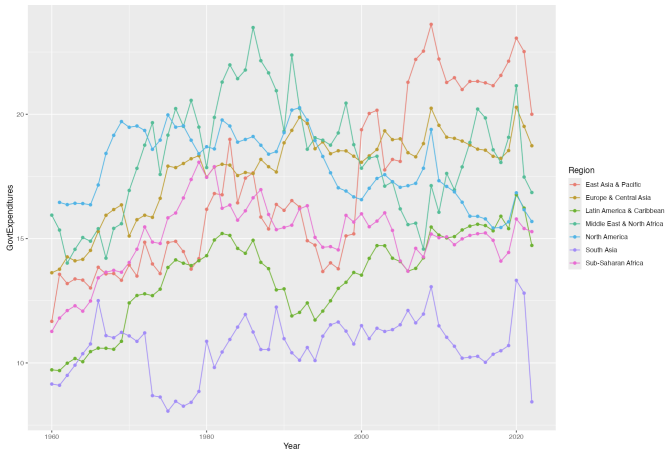
By Group Time Trend Graph

Select variable to
display

GovtExpenditures▼

Select variable to
group by

Region▼



Fancy Scatterplots:

Scatter Plot

Select the x variable to display

InGovtExp

Select the y variable to display

InInflation

Select the variable to be reflected by dot size

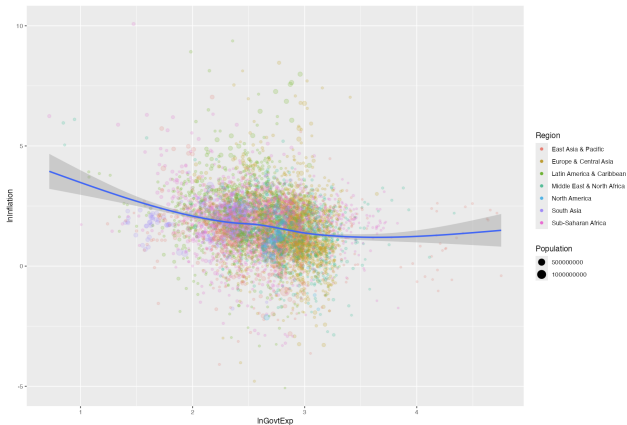
Population

Select the variable to be reflected by color

Region

☐ Sample 1,000 observations to display if number of observations is higher

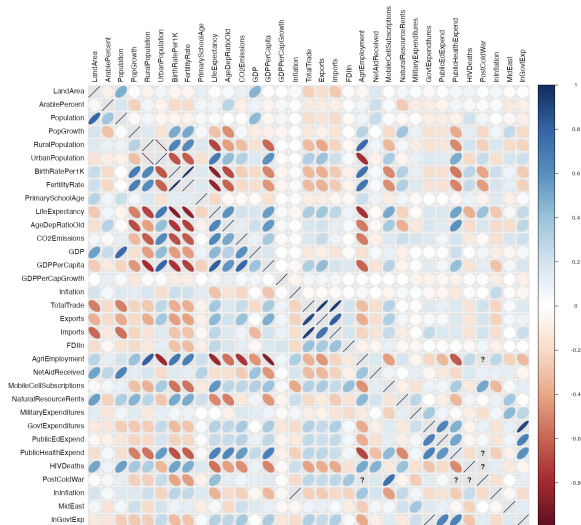
☒ Display smoother



Bivariate Correlations:

Correlation Plot

This plot visualizes sample correlations (Pearson above, Spearman below diagonal). Reports correlations for all continuous variables. Hover over ellipse to get rho, P-Value and n.



Regression (OLS) Analysis:

Regression Analysis

Select the dependent variable

InInflation ▼

Select independent variable(s)

InGovtExp
PopGrowth
PostColdWar
TotalTrade
FDIIn

<i>Dependent variable:</i>	
	InInflation
InGovtExp	-0.597*** (0.039)
PopGrowth	-0.006 (0.010)
PostColdWar	-0.670*** (0.036)
TotalTrade	-0.004*** (0.0003)
FDIIn	-0.002*** (0.001)
Constant	4.090*** (0.113)
Estimator	ols
Fixed effects	None
Std. errors clustered	No
Observations	6,100
R ²	0.141
Adjusted R ²	0.140
Note: *p<0.1; **p<0.05; ***p<0.01	

- Tuesday, June 23: One- and Two-Way “Unit Effects” Models (fixed, “random,” etc.)
- Wednesday, June 24: Dynamics in Panel Data
- Thursday, June 25: Panel Data and Causal Inference
- Friday, June 26: Models for Discrete Responses (+ Exam)